

MP363 – Quantum Mechanics I

Problem Set 1

Due by the end of lecture on Friday, 16 October 2015

1. The Planck blackbody formula says that the spectral radiance per unit frequency emitted by a blackbody at temperature T is

$$B(\nu, T) = \frac{2h}{c^2} \frac{\nu^3}{e^{h\nu/k_B T} - 1}.$$

Rayleigh and Jeans computed the same quantity using only classical methods and obtained the Rayleigh-Jeans (RJ) formula

$$B_{\text{RJ}}(\nu, T) = \frac{2k_B T}{c^2} \nu^2.$$

Show that the first two nonvanishing terms in the Taylor series expansion (around $\nu = 0$) of $B(\nu, T)$ are

$$B(\nu, T) = \frac{2k_B T}{c^2} \nu^2 - \frac{h}{c^2} \nu^3 + \dots$$

and thus explain why Planck's formula agrees closely with the RJ formula for frequencies much less than $k_B T/h$.

(**Hint:** it will be somewhat easier if you first find the first two terms in the Taylor series expansion of

$$\frac{\nu}{e^{h\nu/k_B T} - 1}$$

and then multiply your result by $2h\nu^2/c^2$ to get $B(\nu, T)$. You may have to use l'Hôpital's rule a few times to get this Taylor series.)

2. The colour and temperature of a radiating blackbody are related by *Wein's displacement law*, which states that, for a blackbody at temperature T , the majority of the radiation emitted has wavelengths at or near λ_W given by

$$\lambda_W T = 2.9 \times 10^{-3} \text{ m} \cdot \text{K}.$$

We want to derive and use this relation in the following way:

- (a) First, use the relation $\nu = c/\lambda$ to determine the function $f(\lambda, T)$ such that

$$B(\nu, T) d\nu = -f(\lambda, T) d\lambda.$$

where $B(\nu, T)$ is Planck's blackbody formula given above. This new function $f(\lambda, T)$ gives the spectral radiance due to radiation with wavelengths between λ and $\lambda + d\lambda$.

- (b) Let λ_W be the wavelength for which $f(\lambda, T)$ is maximised (for a fixed temperature); thus, the majority of emitted energy has this wavelength. Show that, if x is the dimensionless number $hc/\lambda_W k_B T$, then x satisfies

$$x = 5(1 - e^{-x}).$$

- (c) The only positive solution to the above equation is $x \approx 4.965$. From this, obtain Wein's displacement law. (Use the values $h = 6.63 \times 10^{-34} \text{ J} \cdot \text{s}$, $c = 3.00 \times 10^8 \text{ m} \cdot \text{s}^{-1}$ and $k_B = 1.38 \times 10^{-23} \text{ J} \cdot \text{K}^{-1}$.)
- (d) Believe it or not, the surfaces of most stars behave very much like blackbodies. Use this fact and Wein's displacement law to predict the colour of a star with surface temperature 5500 K. (It's up to you to scour the library/Internet/whatever to find the relationship between wavelength and colour.)
3. When electromagnetic radiation of frequency $\nu = 2 \times 10^{15} \text{ Hz}$ shines on a sample of gold, the emitted electrons have a kinetic energy of 3.0 eV. When the frequency is increased to 10^{16} Hz , the electron kinetic energy is 36.1 eV.

- (a) Use these data and the photoelectric energy conservation equation

$$h\nu = \frac{1}{2}mv^2 + W$$

to estimate both the gold work function W_{Au} (in eV) and Planck's constant h (in J·s).

- (b) The work function of potassium is 2.3 eV; use your result for h from (a) to estimate the minimum frequency necessary to induce photoelectric emission from a potassium sample.

(Remember that 1 eV = $1.6 \times 10^{-19} \text{ J}$.)

4. A hydrogen atom consists of an electron of charge $-e$ and mass m_e and a proton of charge $+e$ and mass m_p (which is 1836 times larger than m_e). Bohr's model – which assumes that orbital angular momentum is quantised in integer chunks of $h/2\pi$ (i.e. $\hbar$) – gives the energy levels as

$$E_n = -\frac{\mu e^4}{8\epsilon_0^2 h^2} \frac{1}{n^2}$$

where $n = 1, 2, \dots$ and $\mu = m_e m_p / (m_e + m_p)$ is the reduced mass of the electron-proton system.

- (a) Muonium is a hydrogen-like “atom” where an electron orbits an antimuon with charge $+e$. Given that the antimuon mass is 207 times larger than m_e , describe how the energy levels of muonium compare with those of hydrogen.

- (b) Positronium consists of an electron and a positron orbiting around each other. The positron has charge $+e$ and its mass is the same as the electron's; compare its energy levels with those of hydrogen.